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Task NO: 02

Q NO: 01

Q1. Design a C++ program to create a class named Employee. The class should contain the following **private data members**:

- id — to store the employee ID
- name — to store the employee name
- salary — to store the employee salary

Create a **public member function** named setData() that takes the employee ID, name, and salary as arguments and stores them in the private data members.

In the main() function:

1. Ask the user how many employees they want to enter.
2. Use a **for loop** to repeatedly take data for each employee.
3. Create an object of the Employee class for each employee.
4. Call the setData() function to store the entered information.

```
#include <iostream>
```

```
using namespace std;
```

```
class Employee
{
private:
    int id;

    string name;

    double salary;

public:
    void setData(int employeeID, string employeeName, double employeeSalary)
    {
        id = employeeID;
        name = employeeName;
        salary = employeeSalary;
    }

    void display()
    {
        cout << "ID: " << id << endl;
        cout << "Name: " << name << endl;
        cout << "Salary: " << salary << endl;
    }
};

int main()
{
```

```
int n;
```

```
cout << "Enter number of employees: ";
```

```
cin >> n;
```

```
for (int i = 0; i < n; i++)
```

```
{
```

```
    Employee emp;
```

```
    int id;
```

```
    string name;
```

```
    double salary;
```

```
    cout << "\nEnter Employee " << i + 1 << " data:" << endl;
```

```
    cout << "Enter ID: ";
```

```
    cin >> id;
```

```
    cout << "Enter Name: ";
```

```
    cin >> name;
```

```
    cout << "Enter Salary: ";
```

```
    cin >> salary;
```

```
    emp.setData(id, name, salary);
```

```

        cout << "\nEmployee " << i + 1 << ":" << endl;

        emp.display();

    }

    return 0;
}

```

Q NO: 02

Design a C++ program to create a class named Book. The class should contain the following **private data members**:

- bookID — to store the book ID
- title — to store the book title
- author — to store the author's name

Create a **public member function** named setData() to store book information. Use **function overloading** to create two versions of the setData() function:

- One function should accept bookID, title, and author.
- The overloaded function should accept only bookID and title, and set the author name to "Unknown".

In the main() function:

1. Ask the user how many books they want to enter.
2. Use a **for loop** to repeatedly take data for each book.
3. Create an object of the Book class for each book.
4. Ask the user whether they want to enter the **author name** or not.
5. Call the appropriate overloaded setData() function based on the user's choice.
6. Store the information using the selected function.

Concepts covered: Classes, Objects, Private Data Members, Public Member Functions, Access Modifiers, for Loop, User Input, and **Function Overloading**.

```
#include <iostream>
```

```
#include <string>

using namespace std;

class Book
{
private:
    int bookID;
    string title;
    string author;

public:

    // setData with author
    void setData(int id, string bookTitle, string bookAuthor)
    {
        bookID = id;
        title = bookTitle;
        author = bookAuthor;
    }

    // Overloaded setData without author
    void setData(int id, string bookTitle)
    {
        bookID = id;
        title = bookTitle;
        author = "Unknown";
    }
}
```

```
}
```

```
void display()
```

```
{
```

```
    cout << "Book ID: " << bookID << endl;
```

```
    cout << "Title: " << title << endl;
```

```
    cout << "Author: " << author << endl;
```

```
}
```

```
};
```

```
int main()
```

```
{
```

```
    int n;
```

```
    cout << "Enter number of books: ";
```

```
    cin >> n;
```

```
    for (int i = 0; i < n; i++)
```

```
    {
```

```
        Book book;
```

```
        int id;
```

```
        string title;
```

```
        string author;
```

```
        char choice;
```

```
cout << "\nEnter Book " << i + 1 << " data:" << endl;

cout << "Enter Book ID: ";

cin >> id;

cout << "Enter Title: ";

cin >> title;

cout << "Do you want to enter author name? (Y/N): ";

cin >> choice;

if (choice == 'Y' || choice == 'y')
{
    cout << "Enter Author: ";

    cin >> author;

    book.setData(id, title, author);
}
else
{
    book.setData(id, title);
}

cout << "\nBook " << i + 1 << ":" << endl;

book.display();
}
```

```
return 0;  
}
```